

INSTITUTE OF ACTUARIES OF INDIA

Subject SP6 - Financial Derivatives

November 2024 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:

The maximum value of a call option is the value of the underlying itself. This is because the call is an option to buy an asset and hence its value cannot be more than the price of the asset.

For the minimum values:

- The value has to be greater than 0
- If we construct a portfolio of one call option, short share and bond with price $X/(1+r)$ where X is the strike price, r is interest rate and S is current share price, then at expiry, we would have

	Current Value	$S(t) \leq X$	$S(t) > X$
Call	$c(0)$	0	$S(t) - X$
Short Sell	$-S(0)$	$-S(t)$	$-S(t)$
Buy Bond	$X/(1+r)$	X	X
Total	$c(0) - S(0) + \{X/((1+r)^T)\}$	$X - S(t)$	0

Hence $c(0) > \max(0, S(0) - X/(1+r))$

Similarly $p(0) > \max(0, X/(1+r) - S(0))$

Initial Bound on American option would be current intrinsic value

$C(0) \geq \max(0, S(0) - X)$

$P(0) \geq \max(0, X - S(0))$

For Call option, European lower bound is higher than American.

We will expect American call option to be worth more than European call option given additional feature of any time exercise. Thus lower bound of European call option holds for the American option.

Hence,

$C(0) > \max(0, S(0) - X/(1+r))$

European put is lower than American put hence initial bound is correct.

[5]

Solution 2:

The annual marginal PD is

$$d = \frac{1.5\%}{(1 - 0.50)} = 3.00\%$$

Hence, the cumulative PD for five years is

$$d + s_1d + s_2d + s_3d + s_4d = 3.00\% (1 + 0.970 + 0.941 + 0.913 + 0.885) = 14.1\%$$

Where

$$s_1 = (1 - 3.00\%) = 0.970$$

$$s_2 = s_1(1 - 3.00\%) = 0.941$$

$$s_3 = s_2(1 - 3.00\%) = 0.913$$

$$s_4 = s_3(1 - 3.00\%) = 0.885$$

The expected number of defaults is thus $100 * 14.1\%$ or approximately 14. With a recovery rate of 50%, the expected loss is therefore 7% of the notional. Thus, all tranches up to the 7% point are wiped out.

[5]

Solution 3:

- i. By hedging in an appropriate way using futures the farmer may be able to protect himself from fluctuations in the price of barley in 12 months.

An example of such a hedge may be by shorting some suitable barley futures. In this way if the price of barley falls, compared to the current price of 7.50, then he would make less income from the contract with the distillery, but he would make a gain on the short futures he had purchased. In the scenario where the price of barley increases, compared to the current price of 7.50, he would make more income from the contract with the distillery, but he would make a loss on the short futures position.

In general, hedging using futures does not improve the financial outcome of the contract in 12 months. Instead suitable hedging using futures can make the financial outcome of the contract in 12 months more certain.

Basis risk:

Barley futures are available on many exchanges. This may mean the basis risk is relatively low, but it will depend on:

- The contract size of the available futures: If the barley future contracts are much larger, then there will be significant basis risk. In case of the barley futures contracts are smaller, then it would require multiple futures to be bought to mitigate this risk.
- The liquidity of the market for barley futures. Greater liquidity will reduce the basis risk, particularly if the farmer has to close out its future's position early.
- The maturity dates of the futures relative to the distillery contract as it may be the case that the farmer has to close out a futures position well before the maturity of the future. Any rollover risk, this is likely to be low given the hedging period is only going to be a maximum of 12 months.
- The underlying for barley futures (quality/grade etc.) maybe different from the barley that has been contracted and the effectiveness of the hedge depends on the correlation between the prices of the two grades remaining intact.
- Any volatility in the risk-free risk during the contract period can also increase basis risk.

Operational risk:

- The farmer will require sufficient liquidity to meet any mark to market related margin payments over the period of the hedge.
- There is likely to be limited counter-party risk using futures as an exchange is involved. The farmer has no experience in using derivatives which may increase the risk of the hedge been set-up incorrectly.
- The volume of futures contract that minimises risk in the hedge is difficult to determine and may be beyond the skills of the farmer and his

employees. In addition, due to the lack of experience the farmer may require third party advice which will mean additional costs to the hedging.

Thus, overall, the use of futures is likely to increase the certainty of the financial position in 12 months. However, it will not completely eliminate the risk of a fall in the spot price of barley in 12 months.

(5)

- ii. Assume the farmer is able to satisfy the contract and receives full payment. Let w be the breakeven market price of barley in 12 months. In 12 months the cashflow from the contract is $4,000 \times w$.

Discounting to the current time gives a present value of $4,000 \times w \times e^{-0.04}$.

Solving the equation of value: $4,000 \times w \times e^{-0.04} = 25,000$ gives $w = 6.51$.

(2)

iii.

- a. The cost to set-up the hedge is $-4,000 \times 0.37 + 4,000 \times 0.02$.

This is equal to an initial income of 1,400, so there is no cost.

It is assumed there are no other fees relating to the purchase/sale of the derivatives, for example: transaction costs, margin funding, consulting fees; or bid/offer spreads relating to the market prices of the options. (3)

- b. Assume no transaction costs or taxes. Let w be the market price of barley in 12 months where the farm makes neither a profit nor loss. In 12 months the cashflow from the contract is $4,000 \times w$.

Discounting to the current time gives a present value of $4,000 \times w \times e^{-0.04}$.

The cost at the current time of setting up the hedge is -1,400 (using the previous part) and the present value of production is an outflow of 25,000.

In 12 months the cashflows from the hedge will be:

- $4,000 \times \max(6 - w, 0)$ for the put options; and
- $4,000 \times \min(w - 8, 0)$ for the call options.

Discounting this to the current time and combining with the cashflows above gives a value of:

$$4,000 w e^{-0.04} + 1,400 - 25,000 + 4,000 e^{-0.04} (\max(6 - w, 0) + \min(8 - w, 0)) = 0$$

$$\text{Rearranging: } w + \max(6 - w, 0) + \min(8 - w, 0) = 6.14078$$

To solve this equation consider the following cases:

Case 1: $w < 6$

The equation becomes: $w + (6 - w) + (0) = 6.14078$.

This has no solution.

Case 2: $w > 8$

The equation becomes: $w + (0) + (8 - w) = 6.14078$.

This also has no solution.

Case 3: $6 \leq w \leq 8$

The equation becomes: $w + (0) + (0) = 6.14078$.

This has the unique solution: $w = 6.14078$.

(5)

- c. Using this hedge the market price of barley at which the whole contract becomes profitable is 6.14078 compared to an unhedged market price of 6.51. This means that there is some profitable protection against a fall in the market price of barley over the next 12 months. This is exactly the risk from which the farmer was seeking protection.

Due to the use of options the amount of profit is capped above 8, so upside potential is limited. Similarly, deep losses are also reduced if the market price of barley falls below 6.

This hedge does not protect against any changes to the cost of production over the period; For example, the 25,000 cost that has been assumed as constant here may in reality change over the year due to rising prices of labour or seed prices.

(2)

[17 Marks]

Solution 4:

- i. Apply the Black Scholes formula given by $C = SN(d_1) - Xe^{-rT}N(d_2)$

Substituting the numbers, this works out to 2.374 (1)

- ii. $\Delta = N(d_1)$

$$\text{Gamma} = \frac{N'(d_1)}{S\sigma\sqrt{T}}, \text{ where } N'(d_1) = \frac{1}{\sqrt{2\pi}} e^{-\frac{d_1^2}{2}}.$$

Substituting the values, we get $\Delta = 0.562$ and $\text{Gamma} = 0.0747$

(4)

- iii. To ensure delta neutrality, he needs to buy shares and thus borrow money. Let us say he buys “n” shares and borrows an amount B. The following two equations need to be satisfied

Portfolio has to be self-financing: $1000(2.374) - n(50) - B = 0$

Portfolio has to be delta neutral: $1000(0.562) - n(1) = 0$

- a. This implies the number of shares to be bought is 562 (2)

- b. The amount of money to be borrowed is this 25,726 (2)

- iv. If the stock price moves to 52, the delta will have to be recalculated with 64 days as time to expiry. The new delta for the updated price is 0.7017.

This implies that the total number of shares required to be delta neutral on this day is 701.7 or approximately 702.

Thus, the writer needs to buy an additional 140 shares to remain delta neutral. (3)

- v. To ensure delta neutrality, he needs to buy shares, options and thus borrow money. Let us say he buys “ n_1 ” shares, “ n_2 ” options and borrows an amount B. The following two equations need to be satisfied

Portfolio has to be self-financing: $1000(2.374) - n_1(50) - n_2(1.1466) - B = 0$

Portfolio has to be delta neutral: $1000(0.562) - n_1(1) - n_2(0.2965) = 0$

Portfolio has to be gamma neutral: $1000(0.0747) - n_2(0.0529) = 0$

- a. This implies that the number of new options (option 2) that he has to buy is 1412.10 or approximately 1412. (2)
- b. This implies the number of shares to be bought is 143.31 or approximately 143. (2)
- c. The amount of money to be borrowed is this 6410.75. (2)
- d. The reason the amount of money required is less is because of the presence of a large number of options where the current outflow is the premium only. (2)

[20]

Solution 5:

- i. Monte Carlo simulation requires the application of risk neutral pricing of underlying liability by using the principle of market consistency and arbitrage free pricing. Future risk neutral stochastic projection of interest rates, equity returns, corporate bond and property returns will be required.
 - Interest rate – HJM, LIBOR model
 - Equity – constant volatility or if needed jump diffusion model
 - Property - constant volatility model

These models need to be calibrated to the current market prices to give market consistent pricing. We need correlation matrix to correlate the random number used in stochastic projection.

- Key inputs required are
 - Interest rate model: current interest rate, swaption prices across different maturities
 - Equity & property: option prices and volatility estimates across different maturities

- We also need historical return index to estimate the correlation between the different asset classes

(4)

ii. Growth fund of the asset would change according to the table below:

		Equity		
Initial Asset Mix		Actual Exposure	50%	-50%
Equity	30%	300	450	150
Corporate Bond	30%	300	300	300
Government Bond	30%	300	300	300
Property	10%	100	100	100
	100%	1000	1150	850
	% change		15%	-15%

We need to apply 15% up and -15% down for the liabilities or policy holder share.

The balance sheet becomes:

Balance sheet at t = 0		Guarantee Delta 12%	
		50% Rise	50% Fall
Growth Fund	1000	1150 (700+300*1.5)	850 (700+300*(-0.5))
Match Fund	600	600	600
Total Asset	1600	1750	1450
Policyholders Share	1200	1380 (1200*1.15)	1020 (1200*0.85)
Guarantee Costs	300	278.40 (300-180*0.12)	321.60 (300+180*0.12)
Total Liability	1500	1658.40	1341.60
Free Capital	100	91.60	108.40

As can be observed from the above table for delta of 12%, free capital is lower for equity rise. This is due to the fact there are 2 changes in opposite direction.

- Mismatch between asset and policy holder share
- Change in guarantee due to increase in policyholder share

(6)

iii. Having notional positive exposure in growth fund would mean if the property goes down then policyholder share would go down due to link between the % return on growth fund and policyholder share. Over assets will not change as negative in growth fund will be compensated by positive in match fund.

The balance sheet becomes:

Balance sheet at t = 0				
		Notional Property	20% Property Price Rise	20% Property Price Fall
Growth Fund	1000	1100	1140 (900+200*1.2)	1060 (900+200*0.8)
Match Fund	600	500	480 (600-100*1.2)	520 (600-100*0.8)

Total Asset	1600	1600	1620	1580
Policyholders Share	1200	1200	1244 (1200*1140/1100)	1156 (1200*1060/1100)
Guarantee Costs	300	300	294.72 (300-44*0.12)	305.28 (300+44*0.12)
Total Liability	1500	1500	1538.72	1461.28
Free Capital	100	100	81.28	118.72

Suggested strategy does work for XYZ as the free capital will increase in case of a property fall offsetting the fall in the value of their own property. However, this will come at the cost of the policyholder and hence the regulator may not allow this kind of transaction.

(6)

[16]**Solution 6:****i. Reasons for choosing a full yield curve model:**

The rationale for a term structure yield curve model is to be able to price simultaneously options spread across the entire range of maturities in a yield curve.

Exotic interest rate swaps and options include spread options, Bermudan swaptions and path dependent options. These depend not just on the evolution of forward rates along the curve, but on the correlation between changes.

(2)

ii. Desirable features:

- Model is arbitrage free, so produces prices consistent with the market
- Flexible enough to cope properly with the required range of derivative contracts
- Easy to specify and calculate
- The current swap (or bond) curve should be reproduced by the model
- Easy to calibrate
- Enough degrees of freedom (parameters) to make the model flexible to cope with any yield curve shape, but not overly flexible so there is instability between parameters from one day to the next
- Volatility of rates of different maturity should be different, with shorter rates usually seeing higher volatility
- Imperfect correlation between forward rates
- Negative interest rates should not normally be allowed
- Mean reversion of interest rates
- Reasonable dispersion of rates over time

(4)

iii. No – arbitrage models:

No-arbitrage, or “arbitrage free”, models are a class of models that allow recovery of market prices of one set of securities given prices of another set. This creates a world of “relative” prices.

In a non-“arbitrage free” model, securities could be priced using the model and then traded at a different price in the real world, leading to persistent profit – something not tenable in the financial markets.

No-arbitrage is very important in yield curve models, since most complex structures are limiting cases of simpler structures (such as swaps, caps, floors) and ideally the model should recover the prices of the latter exactly.

Also, hedging is done using the simpler structures, so the accounting process will not be distorted by imaginary gains and losses.

(2)

- iv. A trinomial tree is a discrete-time representation of the continuous rate process, with the stochastic process defined by branches which can have three states: up, down and mid-way between.

If the probabilities of going up, down and middle are p_u , p_d and p_m respectively, then $(p_u + p_d + p_m) = 1$ and all values are strictly positive.

The time horizon is split into constant intervals of time Δt , so the rates are compounded Δt period rates.

Further, accuracy is improved with smaller time-steps, up to a point.

The tree for the one-factor model is built in two stages.

Stage 1:

First, set $b(t)$ to zero. Then,

$$dr^* = -ar^*dt + \sigma dz$$

which gives a rate process r^* that is initially zero, and whose evolution is governed by a constant mean reversion towards zero.

The value of σ is determined from the caplet volatilities.

There are three unknowns: p_u , p_d and Δr^* (since $p_m = 1 - p_u - p_d$).

Δr^* can be set at $\sigma\sqrt{3\Delta t}$.

The other two equations come from the expectation and variance of Δr^* over the interval Δt , in which the tree must match the process.

There are three different forms of the mean and variance equations for the tree: when all the branches point upwards or horizontal, when all point downwards or horizontal, and when they straddle the line of unchanged rates.

There will be different probabilities for each situation.

Since $E(\Delta r^*)^2$ contains the term $\sigma \Delta t$, the equations account for the volatility component.

Stage 2:

Now add back in the term structure, to move back from r^* to r .

Let $\alpha(t) = r(t) - r^*(t)$. Then:

$$d\alpha = a[b(t) - \alpha(t)]dt$$

which is a deterministic mean-reverting process (i.e. varies over time but not stochastically), which can be calibrated to the current term structure.

(7)

v. Does HW fulfil the desirable features?

The model is a suitable no-arbitrage full yield curve model as

- It reproduces vanilla bond and swap prices exactly and prices options on these
- The model incorporates mean reversion and time-dependent parameters, so is flexible, behaves well and has little chance of producing negative rates
- It is relatively easy to specify mathematically, though some care is required.

However, a few issues that need to be mentioned are

- It only has one driving factor, so all forward rate moves have to be completely correlated and hence it cannot re-create complex yield curve changes
- The underlying distribution of the interest rate is normal, not log-normal, so it can lead to negative rates [theoretically, as in practice the probabilities of negative rates are negligible]
- The model may be hard to calibrate to cap prices which are priced in the market with the Black's log-normal model.

Also,

- Differences arise as interest rates move away from the current level. These effects, if not disentangled, will lead HW to have an incorrect balance between its volatility and mean-reversion parameters.
- Using constant values for a and σ limits the shapes of yield curve that can be modelled. Time-dependent mean reversion and volatility parameters can fit any current yield curve and forward volatility "hump" shape although they can imply an implausible evolution of the term structure of volatility.

(5)

[20]

Solution 7:

- i. The Black-Scholes pricing model does not explicitly consider drift (the expected return of the underlying asset) because it operates under the risk-neutral valuation principle. In the risk-neutral world, all investors are indifferent to risk. This is a hypothetical world where investors get only risk-free rate of return on their investments. The risk-neutral valuation assumes that the underlying asset grows at the risk-free rate rather than its actual expected return (drift). (4)
- ii. The CIO's comment is correct.
- The Black-Scholes model uses the risk-free rate instead of the actual drift (expected return) of the underlying asset to price options. If a security has a high positive drift the actual expected return can be much higher than the risk-free rate.
 - For short-term options, the impact of drift is minimal because there is less time for the difference between the risk-free rate and the actual drift to compound. However, for long-term options, this difference can become significant. Over a longer period, the positive drift means the underlying asset is more likely to reach higher prices, increasing the intrinsic value and time value of the call option. Hence, the Black-Scholes model, which does not account for this higher expected return, can underprice long-term call options.
 - Hence, if the underlying asset has a higher expected return (positive drift), the probability that the asset price will exceed the strike price at expiration increases. This makes the call option more valuable in reality than what the Black-Scholes model predicts, leading to underpricing by the model.
 - Long dated options are more illiquid and illiquidity premium may also be there which will make the option more expensive. BS model assumes perfect market with unlimited buying and selling without any price impact and doesn't consider illiquidity or associated illiquidity premium. (7)
- iii. The sellers who are aware of the limitations of the Black-Scholes model can adjust their pricing models accordingly.
- Use higher volatility assumption that can compensate them for this model risk.
 - Monte Carlo methods involve simulating the random paths of the underlying asset price using a large number of scenarios. These simulations can accommodate complex features such as path dependency, varying volatility, etc.
 - Use stochastic volatility models, which allows volatility to be a random process itself, rather than a constant. This ensures that a larger universe of scenarios is covered when arriving at the price. In this model, the volatility can be correlated to the stock price.
 - Multi-step binomial trees can also be used to address the issue as they are more flexible.

- Use jump diffusion models that can incorporate sudden jumps in the price of the underlying asset, in addition to the continuous price changes modelled by Black-Scholes.
- Explicit loading for illiquidity risk if it is going to be a negotiated deal if the volatility assumption is not adjusted.

(6)

[17]
